

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281813

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Charnitski; Michael et al.

MOVABLE AND STACKABLE BASE FOR EXERCISE EQUIPMENT

Abstract

The present invention discloses a movable and stackable base system designed for vertical fitness machines. Traditional fitness machines occupy significant floor space and lack the ability to be efficiently stored or moved. This invention addresses these limitations by introducing a base system that enables easy mobility and stacking or nesting of fitness machines, thus optimizing space utilization in gym environments. The system incorporates unique design principles, including lifting mechanisms, specialized base shapes, angled configurations, wheel locking mechanisms, and optional battery power, to facilitate versatile usage of fitness equipment.

Inventors: Charnitski; Michael (Huntington Beach, CA), Packman; Gary (Laguna Hills, CA), Charnitski; Daniel (Rancho Santa Margarita, CA), Charnitski; Richard (Mission Viejo, CA), Schieffer; Shane (Superior, CO)

Applicant: Heart Rate Inc. (Santa Ana, CA)

Family ID: 96948484

Appl. No.: 19/072779

Filed: March 06, 2025

Related U.S. Application Data

us-provisional-application US 63562180 20240306

Publication Classification

Int. Cl.: A63B71/00 (20060101)

U.S. Cl.:

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] This invention pertains to the field of fitness equipment, particularly to the design and functionality of bases for vertical fitness machines. However, one will recognize the bases described herein may apply to any area where equipment needs to be moved and stored in a minimum amount of area. The innovative base system enables improved mobility and storage options for such machines, leading to enhanced space efficiency in fitness facilities in particular. More specifically, this invention relates to bases which may stack or nest with each other to reduce the area needed to store a plurality of the devices. The invention also relates to bases that are equipped with wheels or retractable wheels or other devices to allow the devices to be moved more easily.

Background Art

[0002] Conventional fitness machines are typically large, heavy, and static, occupying substantial floor space in gyms. Examples of such machines are vertical climbing machines, such as VersaClimber brand machines. Such exemplary machines are shown in U.S. Pat. No. 5,492,515, which is incorporated herein by reference. The machine shown in the referenced patent shows a fixed base with no wheels or other structure to aid in mobility. This lack of mobility restricts the dynamic use of fitness spaces and hinders efficient storage solutions. One skilled in the art will note that the base disclosed in the referenced patent, with an extending front member or arm and two extending side members or arms present all in the same plane to form a base to support the machine. This arrangement of extending arms and being in a single plane makes it difficult to efficiently position the machines together for storage. Further, the base of one machine does not allow the base of another machine to be easily placed above or below it for storage.

[0003] The base system proposed in the present application addresses these challenges by allowing fitness machines to be easily moved and stacked, nested, or otherwise positioned together when not in use, thereby minimizing the storage footprint of a number of machines and maximizing floor space utilization.

[0004] The movable and stackable base system offers several advantages over traditional fitness equipment:

[0005] Enhanced space efficiency through compact storage and nesting capabilities.

[0006] Improved mobility for dynamic usage and easy relocation of machines. Versatile deployment options with optional battery power for cord-free operation.

[0007] Enhanced user experience with smooth navigation and user-friendly design elements.

[0008] The movable and stackable base system represents a significant advancement in fitness equipment design, enabling efficient space utilization and enhanced mobility in gym environments. By incorporating innovative features and design principles, the system offers versatile solutions for storage, deployment, and operation of vertical fitness machines, ultimately enhancing the user experience and optimizing fitness facility layouts.

SUMMARY OF THE INVENTION

[0009] The movable and stackable base system for vertical fitness machines revolutionizes the design and functionality of fitness equipment bases. By implementing innovative features such as lifting mechanisms, specialized base shapes, angled configurations, wheel locking mechanisms, and optional battery power, the system enables seamless mobility and nesting of fitness machines. This enhances the versatility and efficiency of fitness spaces, promoting dynamic usage and

optimized storage solutions.

[0010] In operation, the base system facilitates seamless transition between storage and deployment modes:

[0011] Storage Scenario: Machines are lifted slightly, wheels are unlocked, and the battery (if applicable) is switched off. Machines are then nested together, utilizing the specialized base shapes to minimize footprint.

[0012] Deployment Scenario: Machines are easily maneuvered into desired positions for use, with wheels securely locked in place and battery (if applicable) activated for operation.

[0013] In some embodiments, the wheels may be raised or lowered to prevent or allow movement of the base by the wheels.

[0014] At least one wheel preferably swivels to enable easier steering.

[0015] Preferably large wheels, at least 2 inches in diameter, may be employed to provide a smooth user experience, preventing the machine from getting caught on flooring, thresholds, or carpet transitions. Additionally, large wheels ensure a smooth operation on various surface types.

Structural Considerations:

[0016] When lifting the machine, it is crucial to ensure the creation of a rigid base to provide a stable experience. If a V-shape is utilized, additional bracing is necessary at the to prevent flexing or torquing of the members. The bracing may take the form of a cross-member joining or bridging the side arms of the V-shape. The cross-member may take many forms, including plate, angle, or tube stock by way of non-limiting examples. Angle and tube stock may provide superior resistance to flex and torsional forces as well as providing additional support for the user and exercise equipment, particularly when the equipment is being used.

[0017] Preferably the height of the base is lower at the front than it is in the rear section, and particularly lower than the position of the cross member near the middle of the base. This is so the front portion of the base may slide under the cross member of a second base when the bases are in a nested configuration.

[0018] Three wheels (or base contact points) are employed to prevent situations where an uneven floor or manufacturing imperfections cause wobbling. The wheels are preferably located at or near the ends of the arms to provide a more stable arrangement during movement or during use. It is preferred that at least one wheel is mounted to allow swiveling. If only one wheel is mounted to swivel, it is preferred that it be the front wheel or the wheel on the front arm. In some embodiments, the side wheels may pivot to allow improved clearance for stacking or nesting when the side wheels or housing come into contact with the sides of the receiving base being nested into.

[0019] Ideally, a soft compound is utilized for the wheels to prevent slipping during use.

Alternatively, the wheels can be designed to retract, lifting the machine off the floor and resting it on more stable elements.

[0020] Disclosed herein is an arrangement for fitness equipment and bases to provide a nesting capability whereby the machines, when stored, require less than 30% of the machine's length, where:

[0021] To accommodate the storage of multiple machines, angles are utilized (offset from horizontal when viewed from the profile and offset from the vertical when viewed from the top) to ensure no part of the fitness machines prevents a second machine from nesting closely to the first.

[0022] The base of a first machine is configured to slide under the base of a second machine to provide easy and compact storage. The front angle of the machine's base is less than 40-degrees (20-25 degrees in an ideal embodiment) ensuring that the cross-support, such as the front cross-member of the base described in this application, is under the vertical weight-bearing portion of the equipment and is as short as possible thus providing minimal flex and movement of the exercise machine when being used

[0023] A mount with a triangular fin is used for vertical stability of the machine. This fin being of minimal base length to ensure a second machine can nest closely with the first. The ideal

embodiment with the apex of the fin to be between 35 and 40 degrees.

[0024] When seeking to achieve a product offering where storage of the equipment requires a minimal footprint several design constraints will be faced and must be solved. Redesigning the entire piece of equipment is expensive and inefficient. Therefore a design approach requiring minimal machine adjustments while simultaneously achieving desired storage space efficiency is optimal. There are a few steps that, when combined, enable this result. These are: [0025] a) Move away from a design featuring right angles on two of the three axes. [0026] b) Raise the equipment several inches off of the ground onto a structural base with a flat cross-support. [0027] c) Ensure that the cross-support is as narrow as possible. [0028] d) Employ vertical supports with shallow angles at the apex.

[0029] In regard to a) above:

[0030] In engineering, particularly in structural design, there is a strong emphasis on building structures that are based on right angles. When it comes to fitness equipment, such as weight benches, racks, and exercise machines, there's also a strong tendency to design machines that are plumb, level, and square. Such design provides advantages:

[0031] Safety and Stability: Plumb, level, and square fitness equipment ensures safety during use. Stability is crucial, especially when lifting heavy weights or performing vigorous exercises. Proper alignment prevents wobbling or tipping over, reducing the risk of injury to the user.

[0032] Correct Exercise Form: Properly aligned equipment encourages users to maintain correct exercise form. For example, a level weight bench ensures that users can perform exercises like bench presses or dumbbell rows with the correct body positioning, reducing the risk of strain or injury.

[0033] Balance and Symmetry: Plumb, level, and square equipment promotes balance and symmetry during exercises. Uneven surfaces or misaligned components can lead to uneven weight distribution, affecting the effectiveness of workouts and potentially causing discomfort or injury.

[0034] Ease of Use: Fitness equipment that is plumb, level, and square is easier to use and adjust. Users can set up the equipment quickly and confidently without struggling with uneven surfaces or misalignments. This promotes a smoother workout experience and encourages regular exercise participation.

[0035] Durability and Longevity: Properly aligned fitness equipment tends to be more durable and longer-lasting. Misalignments or structural flaws can lead to premature wear and tear, affecting the equipment's lifespan and requiring more frequent maintenance or replacement.

[0036] Professional Appearance: Just like with buildings, plumb, level, and square fitness equipment looks more professional and appealing. Gyms and fitness centers strive to create an environment that is inviting and conducive to exercise, and well-aligned equipment contributes to this atmosphere.

[0037] User Confidence: Users are more likely to trust and feel confident in using fitness equipment that is plumb, level, and square. Confidence in the equipment's stability and reliability enhances the overall workout experience and encourages continued use.

[0038] However, while traditional designs often prioritize square geometry for stability, equipment designed with oblique angles can still maintain stability by carefully considering the center of gravity and weight distribution. Further, by utilizing oblique angles that deviate from square angles on two of the axes, a piece of fitness equipment can be designed to nest closely with another when storing the equipment during periods of non-use. By incorporating angular adjustments, such as sloping frames or tapered components, the equipment can be arranged in a manner that optimizes space efficiency. These oblique angles allow for a snug fit between individual pieces of equipment, reducing wasted space and enabling compact storage arrangements. This strategic use of oblique angles facilitates a seamless and space-saving storage solution, ensuring that fitness facilities can maximize their usable space while maintaining accessibility to equipment when needed.

[0039] In regard to b) above:

[0040] In the modification of fitness equipment to incorporate a nesting or stackable characteristic, paramount consideration is given to modifications of the base structure. The rationale behind this approach is to avoid the costly and inefficient redesign of the entire equipment piece. Instead, a design strategy that necessitates minimal machine adjustments while concurrently achieving optimal storage space efficiency is pursued. One key step in this process, as previously discussed, involves deviating from right angles on two of the three axes. However, the base of the equipment must also undergo a pivotal modification. By elevating the equipment on a base positioned several inches above the ground, nesting capability can be achieved with minimal alteration to the equipment design. This adjustment is significant as most base structures lie in the horizontal plane, inherently prohibiting overlap between two 3-D objects. Elevating the equipment above the base and supporting it on a thin structural platform, such as sheet steel, allows for cross-supports of the equipment to pass above the supporting beams, bars, or tubes, thereby enabling efficient nesting or stacking.

[0041] In regard to c) above:

[0042] When designing a support structure with a steel sheet spanning a gap, the flex of the sheet is not linearly proportional to the span's width. Instead, it follows a non-linear relationship.

[0043] The flexural behavior of a beam (such as a steel sheet used as a beam) is governed by the Euler-Bernoulli beam theory. According to this theory, the deflection of a beam under a load depends on several factors including the span length, the material properties, the cross-sectional shape, and the loading conditions.

[0044] For a simply supported beam (supported at both ends), the deflection (or flex) at the midpoint of the span can be calculated using the following formula derived from beam theory:

$$[00001] \quad \delta = \frac{5qL^4}{384EI}$$

[0045] Where: [0046] δ is the deflection at the midpoint of the beam [0047] q is the applied load per unit length (load per unit area in the case of a steel sheet) [0048] L is the span length [0049] E is the modulus of elasticity of the material (for steel) [0050] I is the second moment of area of the cross-section of the beam.

[0051] From this formula, one skilled in the art can see that the deflection is proportional to the fourth power of the span length. This means that doubling the span length would result in 16 times more deflection, not just twice as much.

[0052] Therefore, the relationship between the flex of the steel sheet and the span's width is non-linear, and wider spans will indeed create a disproportionately larger amount of flex. This is a critical consideration in engineering design to ensure that the deflection of the structure remains within acceptable limits.

[0053] In regard to d) above:

[0054] Lastly, in the realm of designing fitness equipment for space-efficient storage, particularly for the nesting of two machines together, the incorporation of a triangular fin or support emerges as a practical solution owing to the well-established rigidity associated with triangular designs.

However, in scenarios involving the nesting of multiple identical pieces of equipment, a potential challenge arises: the necessity for stability demands a wide base, while space efficiency dictates a narrow base. The, the ideal embodiment necessitates a narrow range of angles to concurrently satisfy both requirements.

[0055] Strategically, the triangular support is crafted with minimal base length to enable the close nesting of a second machine with the first. This reduction in base length effectively minimizes the space occupied by the support structure, facilitating tighter nesting of the machines. Yet, the triangular shape, while adept at providing stability in the vertical direction, often demands a broad base to prevent equipment tipping during usage or storage, a crucial aspect for user safety and equipment integrity.

[0056] In the preferred embodiment, the triangular support features an apex angle within the range of 35 to 40 degrees. This selection optimizes stability while minimizing the support's footprint. By

striking this balance, the weight distribution of the equipment is effectively managed, accounting for the user's position on the machine and providing ample resistance against tipping forces.

[0057] The movable and stackable base system represents a significant advancement in fitness equipment design, enabling efficient space utilization and enhanced mobility in gym environments. By incorporating innovative features and design principles, the system offers versatile solutions for storage, deployment, and operation of vertical fitness machines, ultimately enhancing the user experience and optimizing fitness facility layouts.

[0058] In one form, the invention is directed to a base for fitness equipment. The base includes: a first side member and a second side member; a first side arm joined to the first side member; a second side arm joined to the second side member; a front cross-member and a rear cross-member; and an equipment mount. The first and second side members are joined at a first position to form a "V" shape at a front portion of the base. The front cross-member joins the first and second side members at a second position. The rear cross-member is positioned at a third position. The equipment mount is in contact with the front cross-member and the rear cross-member. The equipment mount includes a vertical support.

[0059] In one form, the maximum height of the front portion above a ground surface under the front portion is less than the height above the ground surface of the bottom surface of the rear cross member.

[0060] In one form, the base further includes a first wheel located at an apex of the front portion of the base, a second wheel located at an end of the first side arm, and a third wheel located at an end of the second side arm. The first wheel is a swivel wheel. The second or third wheel is a lockable wheel.

[0061] In one form, the first and second side members are oriented at an angle of 20 degrees to one another.

[0062] In one form, the base further includes a flange that extends upwards from the front cross-member.

[0063] In one form, the base further includes a second flange that extends upwards from the equipment mount. The second flange is connected to the first flange.

[0064] In one form, the equipment mount further includes a ridge flange.

[0065] In one form, the ridge flange is oriented at an angle of 75 degrees to the top surface of the rear cross-member.

[0066] In one form, the first side arm and the second side arm are oriented 120 degrees from one another.

[0067] In one form, the ridge flange is supported towards the front of the base by the vertical support extending between a surface of the ridge flange and the second flange.

[0068] In one form, the first side arm and first side member are joined at the rear cross-member and the second side arm and second side member are joined at the rear cross-member.

[0069] In one form, a height of the first side arm is greater at the front portion of the base than at the second position.

[0070] In one form, the base is provided in combination with fitness equipment. The center of gravity of the assembled combination is located between the second and third position.

[0071] In one form, the front portion of the base of a first base is positioned under the rear cross member of a second base when the first base and second base are in a nested arrangement.

[0072] In one form, the front portion of the base of a first base rolls under the rear cross-member of a second base when the first base and second base are moved to a nested arrangement.

[0073] In one form, the front portion of the base of a first base is guided by the first and second side members of a second base when the first base and second base are moved to a nested arrangement.

[0074] In one form, the invention is directed to a method of storing fitness equipment. The method includes: providing a plurality of bases for the fitness equipment; and moving a first base of the

plurality of bases from a first location to a second location. Each of the plurality of bases includes: a first side member and a second side member, the first and second side members joined at a first position to form a “V” shape at a front portion of the base; a front cross-member joining the first and second side members at a second position; a first side arm joined to the first side member; a second side arm joined to the second side member; a rear cross-member positioned at a third position; and an equipment mount. The equipment mount is in contact with the front cross-member and the rear cross-member. The equipment mount includes a vertical support having a ridge flange. Fitness equipment is attached to the ridge flange. A first wheel is located at an apex of the front portion of the base, a second wheel is located at an end of the first side arm, and a third wheel is located at an end of the second side arm. The first wheel is a swivel wheel. The second or third wheel is a lockable wheel. The front portion of the first base moves under the rear cross-member of a second base. The front portion of the first base is located within the front portion of the second base in a co-planar relationship.

[0075] In one form, the front portion of the first base is located below the front cross-member of the second base when the first base is in the second location.

[0076] In one form, the invention is directed to a base for fitness equipment. The base includes: a first side member and a second side member, the first and second side members joined at a first position to at a front portion of the base; a front cross-member joining the first and second side members at a second position; a first side arm joined to the first side member; a second side arm joined to the second side member; a rear cross-member positioned at a third position; and an equipment mount. The equipment mount is joined to the front cross-member and the rear cross-member. The equipment mount includes a vertical support having a ridge flange. A vertical climbing fitness machine is mounted to the ridge flange. The center of gravity is located below the ridge flange and between the front cross-member and the rear cross-member.

[0077] In one form, the ridge flange is supported towards the front of the base by the vertical support extending between a surface of the ridge flange and the front cross-member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0078] FIG. **1** is a plan view of nested or stacked bases of the present invention.

[0079] FIG. **2** is a plan view of an alternate base of the present invention.

[0080] FIG. **3** is a rear elevational view of the base of the present invention with an exercise machine attached.

[0081] FIG. **4** is a side elevational view of the base of the present invention with an exercise machine attached.

[0082] FIG. **5** is a top plan view of base of the present invention with an exercise machine attached.

[0083] FIG. **6** is a side elevational view of the equipment mount of the present invention.

[0084] FIG. **7** is a side elevational view of the base of the present invention.

[0085] FIG. **8** is a side elevational view of the base of the present invention.

[0086] FIG. **9** is a top perspective view of the base of the present invention.

[0087] FIG. **10** is a bottom perspective view of the of the base of the present invention.

[0088] FIG. **11** is a front view of the base of the present invention.

[0089] FIG. **12** is a rear view of the base of the present invention.

[0090] FIG. **13** is a top view of the base of the present invention.

[0091] FIG. **14** is a bottom view of the base of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0092] With reference to the figures, in some embodiments the invention includes a base **10** having a front portion **11** and extending arms **13** and **14**. The front portion **11** and extending arms may be

constructed from any materials used to commonly construct bases for exercise equipment. This may include solid, plate, angle, and tubular metal stock. Other materials of sufficient strength may be used.

[0093] In some embodiments, the front portion **11** is constructed in a V-shape. Among other benefits, the v-shape minimizes width while allowing the ability to nest multiple bases for storage, as shown in FIG. **1**. In such an embodiment, the front portion **11** includes two side members **15** and **16**, each side member joined to the other at a joint or vertex **18**. The vertex **18** need not form a hard or sharp angle, but is instead used to describe the general area where the two side members meet or are joined together, preferably by a front joining member **30**. In other embodiments, the side members **15** and **16** may be joined without a joining member. In the most preferred embodiment, the side members **15** and **16** are set at an angle **2** of approximately 20 degrees from each other.

[0094] The extending arms **13** and **14** may be joined to or extend from the respective side members **15** and **16** or from a rear cross-member **20**. The extending arms **13** and **14** may be oriented at an angle of 120 degrees with respect to each other. The rear cross-member **20** may join the side members **15** and **16** in some embodiments. In some embodiments, additional cross-members may be present, such as forward cross-member **22**. The cross-members **20** and **22** are referred to as forward and rearward in relation to each other with respect to the front and rear of the base **10**.

[0095] In some embodiments, cross-members **20** and **22** are positioned to define a central area of the base **10**. The center of gravity CG of the assembled base and exercise equipment is preferably located in the vertical column above the central area of the base. Such placement provides stability of the assembly. The attachment structures for the exercise equipment **1** is preferably in this central area.

[0096] In some embodiments, the cross-members **20** and **22** may include flanges or other structures that extend in a generally upward or vertical direction. The flanges may be used to mount or fasten or otherwise support or attach exercise equipment **1**. The flanges may also be used as a strengthening member. For instance, the rear cross-member **20** may include rear cross-member flange **25** and the front cross-member **22** may include a front cross-member flange **23**. In some embodiments, the rear cross-member flange **25**, the front cross-member flange **22**, and the side arms **15** and **16** define the central area.

[0097] In some embodiments, a front wheel **40** is located at the vertex **18** of the front portion **11**. The front wheel **40** may be mounted to the joining member **30**. The joining member **30** may be positioned on the top surfaces of the side members **15** and **16**. Such positioning allows the front wheel **40** to be positioned in front of the side members **15** and **16** while not increasing the overall height of the front **11** by the height of the front wheel **40**. In some embodiments, the front wheel **40** is positioned in front of the most forward extent of the attached exercise equipment **1**.

[0098] The front wheel **40** preferably swivels. The front wheel **40** may be smaller than other wheels. In some embodiments, the wheels are retractable so that the overall height of the base with respect to the ground is reduced. Such an arrangement may also allow the base **10** to selectively contact the ground or rest on pads to resist or prevent movement of the base with respect to the ground.

[0099] Side wheels **41** and **42** may be located on the outward ends of the extending arms **13** and **14**. The wheels may swivel and may be retractable to allow the base **11** to selectively rest on pads. If the side wheels **41** and **42** do not swivel, they should be oriented in a front to back direction.

[0100] In some embodiments, the front wheel **40** or side wheels **41** and **42** may be retractable, so that the base **10** rests on the ground when the wheels are retracted, and is raised off the ground when the wheels are deployed.

[0101] In some embodiments, such as the alternate embodiment shown in FIG. **2**, the joining member **30'** may join the first side member **15'** and second side member **16'** internal to the V shaped form. The front cross member **22'** may also connect the first and second side members.

With such positioning, the front wheel **40** may be positioned on the joining member **30'** between the first and second side members **15'** and **16'**.

[0102] The rear cross-member **20** may include recesses on the underside to counter-sink fastener heads so that the underside is generally free of obstructions that may snag or interfere with the nesting arrangement of multiple bases when the bases are in a nesting arrangement. That is to say it is preferable to not have obstructions or extensions or protrusions on the underside of any cross-member so that the front portion **11** of one base may slide under the cross-member of another base.

[0103] An equipment mount **60** may be connected to the cross members **20** and **22**. In some embodiments the equipment mount **60** may attach to or be in contact with the top surface of the rear cross-member **20**. In some embodiments, the equipment mount **60** may attach to the rear cross-member flange **25**. In some embodiments, the equipment mount **60** may form the rear cross-member **20**. In some embodiments, the equipment mount **60** may be in contact with the top surface of the front cross-member **22**. In some embodiments, the equipment mount **60** may be attached to the front cross-member flange **23**.

[0104] In some embodiments, the equipment mount **60** may include a triangular support **61** or fin. The triangular support **61** may form an angle **3** of 75 degrees to the top surface of the cross-member in order to support the exercise machine **1**. In the most preferred embodiment, the apex angle **4** of the triangular support **61** is about 37 degrees. In some embodiments, the equipment mount **60** may include flanges. In the most preferred embodiment, the equipment mount **60** includes a ridge flange **61** to support or mount or fasten the spine or back of the exercise machine. The equipment mount **60** may also include a front flange, the front flange **62** restraining the forward movement of the exercise machine **1**. The front flange **62** may be fastened to or stopped by a second cross-member flange **23**, upstanding from the second cross-member **22**.

[0105] Similarly, the equipment mount **60** may include a back flange **63**, the back flange **63** restraining the rearward movement of the exercise machine **1** when the exercise machine is on the base **10**. The back flange **63** may be fastened to or stopped by a rear cross-member flange **25** upstanding from the rear cross-member **20**.

[0106] The base **11** may include a front cross-member **22** located between the rear cross-member **20** and the vertex **18**. The front cross-member **22** joins the side members **15** and **16**. In some embodiments, the equipment mount **60** is connected to the rear cross-member **20** and the front cross-member **22**.

[0107] The front portion **11** has a maximum height off of the ground that is less than the height off of the ground of the bottom of the rear cross-member **20**. This arrangement can be best illustrated in the side views of the drawings, such as FIGS. **4**, **7**, and **8**. The maximum height off the ground of the front portion **11** should not be less than the minimum height off the ground of the extending arms **13** and **14** or side members **15** and **16**. This allows the extending arms **13** and **14** of a first base and side members **15** and **16** of a second or receiving base to guide the front portion **11** of an inserted or first base into a nesting or stacked position for storage, as shown in FIG. **1**.

[0108] In some embodiments it is preferable that the second cross-member **22** be positioned near the location where the base transitions from a first height to the reduced height of the front section **11**.

[0109] The foregoing disclosure of specific embodiments is intended to be illustrative of the broad concepts comprehended by the invention.

Claims

1. A base for fitness equipment, the base including: a first side member and a second side member, the first and second side members joined at a first position to form a V shape at a front portion of the base; a front cross-member joining the first and second side members at a second position; a first side arm joined to the first side member; a second side arm joined to the second side member;

a rear cross-member positioned at a third position; and an equipment mount, the equipment mount in contact with the front cross-member and the rear cross-member, the equipment mount including a vertical support.

2. The base of claim 1, wherein the maximum height of the front portion above a ground surface under the front portion is less than the height above the ground surface of the bottom surface of the rear cross member.

3. The base of claim 1, further including a first wheel located at an apex of the front portion of the base, a second wheel located at an end of the first side arm, and a third wheel located at an end of the second side arm, the first wheel being a swivel wheel, and the second or third wheel being a lockable wheel.

4. The base of claim 1, wherein the first and second side members are oriented at an angle of 20 degrees to one another.

5. The base of claim 1, further including a flange extending upwards from the front cross-member.

6. The base of claim 5, further including a second flange extending upwards from the equipment mount, the second flange connected to the first flange.

7. The base of claim 6, the equipment mount further including a ridge flange.

8. The base of claim 7, wherein the ridge flange is oriented at an angle of 75 degrees to the top surface of the rear cross-member.

9. The base of claim 1, wherein the first side arm and the second side arm are oriented 120 degrees from one another.

10. The base of claim 7, the ridge flange supported towards the front of the base by the vertical support extending between a surface of the ridge flange and the second flange.

11. The base of claim 1, wherein the first side arm and first side member are joined at the rear cross-member and second side arm and second side member are joined at the rear cross-member.

12. The base of claim 1, wherein a height of the first side arm is greater at the front portion of the base than at the second position.

13. The base of claim 1 in combination with fitness equipment, wherein the center of gravity of the assembled combination is located between the second and third position.

14. The base of claim 3, wherein the front portion of the base of a first base is positioned under the rear cross member of a second base when the first base and second base are in a nested arrangement.

15. The base of claim 3, wherein the front portion of the base of a first base rolls under the rear cross-member of a second base when the first base and second base are moved to a nested arrangement.

16. The base of claim 3, wherein the front portion of the base of a first base is guided by the first and second side members of a second base when the first base and second base are moved to a nested arrangement.

17. A method of storing fitness equipment, the method including: providing a plurality of bases for the fitness equipment, each of the plurality of bases including: a first side member and a second side member, the first and second side members joined at a first position to form a V shape at a front portion of the base, a front cross-member joining the first and second side members at a second position, a first side arm joined to the first side member, a second side arm joined to the second side member, a rear cross-member positioned at a third position, an equipment mount, the equipment mount in contact with the front cross-member and the rear cross-member, the equipment mount including a vertical support, the vertical support including a ridge flange, fitness equipment attached to the ridge flange, a first wheel located at an apex of the front portion of the base, a second wheel located at an end of the first side arm, and a third wheel located at an end of the second side arm, the first wheel being a swivel wheel, and the second or third wheel being a lockable wheel; and moving a first base of the plurality of bases from a first location to a second location wherein the front portion of the first base moves under the rear cross-member of a second

base, the front portion of the first base located within the front portion of the second base in a coplanar relationship.

18. The method of claim 17, wherein the front portion of the first base is located below the front cross-member of the second base when the first base is in the second location.

19. A base for fitness equipment, the base including: a first side member and a second side member, the first and second side members joined at a first position to at a front portion of the base; a front cross-member joining the first and second side members at a second position; a first side arm joined to the first side member; a second side arm joined to the second side member; a rear cross-member positioned at a third position; and an equipment mount, the equipment mount joined to the front cross-member and the rear cross-member, the equipment mount including a vertical support, the vertical support including a ridge flange, a vertical climbing fitness machine mounted to the ridge flange, the center of gravity being located below the ridge flange and between the front cross-member and the rear cross-member.

20. The base of claim 19, the ridge flange supported towards the front of the base by the vertical support extending between a surface of the ridge flange and the front cross-member.
